



A personalized design for progressive addition lenses

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Abstract: We propose a numerical method for the design of a progressive addition lens (PAL) that can accommodate more personal needs as compared to using the analytical solution of the Laplace equation. In our method, the auxiliary function $u(x, y)$ of a PAL is obtained by the numerical solution of the Laplace equation with the boundary and link conditions. The boundary condition is obtained using the genetic algorithm with the input from the individual requirement. The link condition is determined using the finite difference method with a smoother $u(x, y)$ on the meridian. Two examples are given for outdoor and office use. In both cases, the astigmatism area is pushed towards a small area near the edge of the lens.

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1. Introduction

A progressive addition lens (PAL) provides seamless clear vision at different viewing distances. There are two main categories of methods for designing PALs. One belongs to direct method. For example, Winthrop *et al.* [1] described a system in which the designers specified the focal power along the umbilical meridian. Both the shape of the remainder of the lens and the curvatures of the progressive surface are determined by the auxiliary function $u(x, y)$. The contours of auxiliary function on x - y plane are called level curves. The auxiliary function was obtained by analytically solving the Laplace equation. Steele *et al.* [2] specified the focal power over the entire surface using conics (as an auxiliary function) and obtained the surface shape of the PAL by solving an elliptic partial differential equation. The other way is to determine the PAL surface indirectly. For example, Loost *et al.* [3], Wang

[4], Wei [5] devised an evaluation function that attempts to reach a balance between the desired distribution of focal power and the unwanted astigmatism. The PAL surface was obtained by numerically minimizing the evaluation function. In the direct methods the designs of the meridian focal power and the level curves are two key points. Recently, the technique on searching the optimized focal power distribution on meridian line has been described [6,7]. Winthrop *et al.* and Steele *et al.* presented the analytic expressions for the level curves [1,2]. All these methods have only two or three parameters to adjust the level curves. Therefore, their capability to meet personal needs for vision correction is limited.

We propose a method that can accommodate more personal needs as compared with the methods mentioned above. In our approach, the level curves are obtained by numerically solving the Laplace equation with the boundary and link conditions that depend on individual situation. There is a complex relationship between the boundary condition of the Laplace equation and the astigmatism. The boundary condition is obtained using the genetic algorithm with the input from the personalized requirement. To minimize the astigmatism on the meridian line, we propose a smoother link condition using variation principle and the finite difference method. The method provides flexibility and efficiency for determination of an individualized lens.

2. Design of the level curves for a progressive addition lens

The surface of a PAL is divided into four regions (Fig. 1). The distance area 1 in the upper portion of the lens has a relatively low focal power. The near area 2 is 10-18 mm below the distance area and has a relatively high focal power. The progressive corridor 3 connects the distance and near areas. The astigmatism areas 4 are at the left and right of the progressive corridor with relatively severe astigmatism. The difference in focal power between the reference point A in the distance area and the reference point B in the near area is regarded as addition power (ADD) of the PAL. The distance area, near area and progressive corridor are called effective vision regions. The astigmatism areas cannot be used to correct the vision of a wearer.

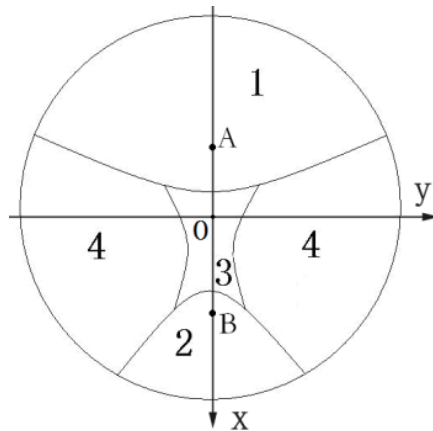


Fig. 1. Four regions of a PAL.

The origin O is the center of the lens and x-y plane is tangent to the lens. The x-axis points downward in the direction of increasing focal power. The z-axis points out of the paper towards the reader. The meridian line connects points A and B. The distance between point A and B is the length of the progressive corridor.

The direct design method is divided into several steps. The first step is to design the meridian focal power (along the meridian line) and the auxiliary function $u(x, y)$. The second

step is to determine the curvature and the centers of curvature at each point on the PAL surface. The last step is to obtain the vector height $z(x, y)$.

The focal power distribution should be smooth over the surface of the lens, so the auxiliary function $u(x, y)$ needs to distribute smoothly. A criterion for smoothness requires that the quadratic sum of the partial derivatives $\partial u / \partial x$ and $\partial u / \partial y$ be a minimum, i.e., the Dirichlet integral is minimum. According to the Euler-Lagrange variation principle, the auxiliary function $u(x, y)$ satisfies the Laplace equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0. \quad (1)$$

We propose to solve Eq. (1) using numerical technique. The boundary condition of the Laplace equation is optimized using the genetic algorithm while the link condition is obtained using the finite difference method.

2.1 The boundary condition of the Laplace equation

The control point u_k represents one of the grid points on the boundary of the computational domain Ω and is defined as

$$u_k = -L + p_k h \quad k = 1, 2, 3, \dots, K \quad (2)$$

Here h is related to the length of the progressive corridor, L is the distance from point A to the original point O, and p_k is the control parameter of the genetic algorithm varying from 0 to 1. K is the number of 'Chromosomes' in the genetic algorithm. The sequence of all 'Chromosomes' p_k constitutes a vector as an 'individual'. The value of u_k varies from $-L$ to $h - L$.

The objective function f of the genetic algorithm meets the merits of the vector [7]

$$f = \alpha_1 |\Phi_1 - \Phi_1^*| + \sum_{i=2}^6 \alpha_i |\bar{\Phi}_i| + \sum_{i=7}^9 \alpha_i |\bar{\Phi}_i - \Phi_i^*|. \quad (3)$$

Here Φ_1 is the maximum astigmatism of the PAL. The maximum astigmatism should meet the requirement $\Phi_1^* = \rho |P_A - P_B|$, where P_A and P_B are the focal powers at points A and B, and ρ is the weighting factor of the additional power. $\bar{\Phi}_i$ ($i = 2, 3, \dots, 6$) are the mean values of the astigmatism in the distance area, near area and progressive corridor and two astigmatism areas respectively. $\bar{\Phi}_i$ ($i = 7, 8, 9$) are the mean power values in the distance area, near area and progressive corridor respectively. Φ_i^* are the corresponding objective values. $\bar{\Phi}_i$ change in the loop of genetic algorithm for searching the optimized boundary conditions. $\alpha_1, \dots, \alpha_6$ are the corresponding areas' weighting factors of the astigmatism. α_7, α_8 and α_9 are the corresponding areas' weighting factors of the focal power difference. ρ ($0.75 \leq \rho \leq 1$) and α_i ($0.1 \leq \alpha_i \leq 2$) are relative values and determined by the preferences of the wearers. For outdoor activities, a wide distance area is needed, so the weighting factor α_2 should be larger than α_3 . For the office activities, a smaller distance area and a larger near area are wanted, so the weighting factor α_3 should be larger than α_2 . In any case, we want the astigmatism as little as possible but the effort is limited by other demand such as the dimensions of clear distance and near regions. Actually, it is a trade-off among the distance area, the near area and the astigmatism.

2.2 The link condition of the Laplace equation

In the previous art [1], the auxiliary function $u(x, y)$ on the meridian line between points A and B is as follows

$$u(x, 0) = x - \quad L \leq x \leq h - L \quad (4)$$

In order to reduce the astigmatism of the PAL, we try to keep the focal power stable beyond point A and point B on the meridian line. The function $u(x, 0)$ should changes more smoothly. At points A and B, $u(x, 0)$ is equal to x , the slopes should be equal to zero, $u(x, 0)$ should have higher order N of the first non-vanishing differential derivatives. On the meridian line between points A and B, the absolute values of the differential derivatives are minimum when the order is less than N or equal to N .

We minimize the summation of the square of the derivatives with the order from 1 to N

$$J[u(x, 0)] = \int_{-L}^{h-L} \left[\sum_{n=1}^N \left(\frac{d^n u(x, 0)}{dx^n} \right)^2 \right] dx. \quad (5)$$

The analytic expression of $u(x, 0)$ for the minimum of Eq. (5) satisfies the Euler-Poisson equation [8]

$$\sum_{n=1}^N (-1)^n \frac{d^{2n} u(x, 0)}{dx^{2n}} = 0 \quad (6)$$

$$\begin{cases} u(-L, 0) = -L \\ u(h-L, 0) = h-L \end{cases} \quad (7)$$

From Eq. (6) and (7) we obtain

$$\left. \frac{d^n u(x, 0)}{dx^n} \right|_{x=-L}^{x=h-L} = 0 \quad n = 1, 2, \dots, (N-1) \quad (8)$$

The characteristic equation of the differential Eq. (8) is

$$(-1)^N m^{2N} + (-1)^{(N-1)} m^{2(N-1)} + \dots + m^4 - m^2 = 0. \quad (9)$$

Equation (6) has two zero solutions and $2N-2$ nonzero ones. Assuming m_i ($i = 1, 2, \dots, 2N-2$) are the nonzero solutions, the general solution of the differential Eq. (6) is

$$u(x, 0) = \sum_{i=1}^{2N-2} C_i e^{m_i x} + C_{2N-1} x + C_{2N} \quad (10)$$

From Eq. (7) and Eq. (8), C_i ($i = 1, 2, \dots, 2N$) in Eq. (10) are obtained. Then the auxiliary function $u(x, 0)$ on meridian line is obtained.

Further, $u_{i,j}$ at two sides of the meridian line with width d is determined by the finite difference scheme [9]. We use a square grid (x_i, y_j) to numerically calculate $u_{i,j}$. Given $u_{i,j} = u(x_i, y_j)$, the centered finite difference formula is used for the second derivative

$$\left(\frac{\partial^2 u}{\partial y^2} \right)_{i,j} = \frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{\Delta y^2}. \quad (11)$$

Here Δy is the step size. Supposing the symmetric axis of $u(x, y)$ is the meridian line, $u_{i,j+1}$ is equal to $u_{i,j-1}$. Rearranging Eq. (11), we obtain

$$u_{i,j\pm 1} = u_{i,j} + \frac{1}{2} \Delta y^2 \left(\frac{\partial^2 u}{\partial y^2} \right)_{i,j} \quad (12)$$

Based on the Laplace equation and add an optimization factor a_u , we obtain

$$u_{i,j\pm 1} = u_{i,j} - \frac{1}{2} a_u \Delta y^2 \left(\frac{\partial^2 u}{\partial x^2} \right)_{i,j} \quad (13)$$

Then the values of $u_{i,j\pm n}$ $n=1,2,3\dots$ are analogized in turn. The values of $u(x, y)$ between the left and right boundaries of the progressive corridor are obtained. The width of the progressive corridor and the optimization factors a_u change according to different personal needs.

2.3 Numerical solution of the Laplace equation

The Laplace equation with the boundary and link conditions obtained above can be written as

$$\begin{cases} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, & (x, y) \in \Omega \\ u(x_\Gamma, y_\Gamma) = \phi(x_\Gamma, y_\Gamma) & (x_\Gamma, y_\Gamma) \in B_\Gamma \\ u(x_L, y_L) = \varphi(x_L, y_L), & (x_L, y_L) \in D_L \end{cases} \quad (14)$$

Here the domain Ω is a square region tangent to the PAL, B_Γ the boundary, D_L the link condition area, $\phi(x_\Gamma, y_\Gamma)$ the optimized boundary condition, and $\varphi(x_L, y_L)$ the link condition.

The Laplace equation is changed into a set of difference equations by the finite difference scheme.

$$\begin{cases} u_{i,j} = \frac{1}{4}(u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}), & 1 \leq i \leq m-1; 1 \leq j \leq m-1 \\ u_{i,j} = \phi(i_\Gamma g, j_\Gamma g), & i_\Gamma = 0, m, \quad 0 \leq j_\Gamma \leq m \\ & j_\Gamma = 0, m \quad 0 \leq i_\Gamma \leq m \\ u_{i,j} = \varphi(i_L g, j_L g), & (i_L g, j_L g) \in D_L \end{cases} \quad (15)$$

Here $g = \Delta x = \Delta y$ is the step and the side length of the square Ω is mg with m an integer. Linear Eqs. (15) are solved by the successive cover-relaxation (SOR) approach [10]. The SOR technique employs a repetitive series of sweeps over the mesh to converge on a solution. The rate of convergence depends on the value of the Over Relaxation Factor (ORF), and a preferred value of the ORF is determined experimentally. An important advantage of the SOR technique is that it reaches convergence in a time proportional to the square root of the number of mesh points. This feature implies that at modest cost in computational time, a sufficient mesh density can be implemented for SOR to converge to the solution.

3. Examples and discussion

We apply the proposed method to two examples to demonstrate how a specific distribution of the focal power and astigmatism of a PAL is achieved by the corresponding boundary and link conditions. In the first example, the wearer uses the PAL for outdoor activities. Therefore, a wide distance area is needed. According to the prescription, the PAL has a -2.00 diopter focal power in the distance area and a $+2.00$ diopter addition power. The index of refraction of the lens material is 1.523. The front surface of the PAL is a spherical surface with $+2.00$ diopter focal power. The back surface is a progressive addition surface with -4.00 diopter focal power in the distance area and -2.00 diopter focal power in the near area. The values of h and L are 34 and 17 respectively.

To compare the performance of the proposed method with the previous analytical methods, a progressive surface is calculated by Winthrop method. The solution of Laplace equation is an analytical expression with parameters h , L , x and y . The level curves are shown in Fig. 2.

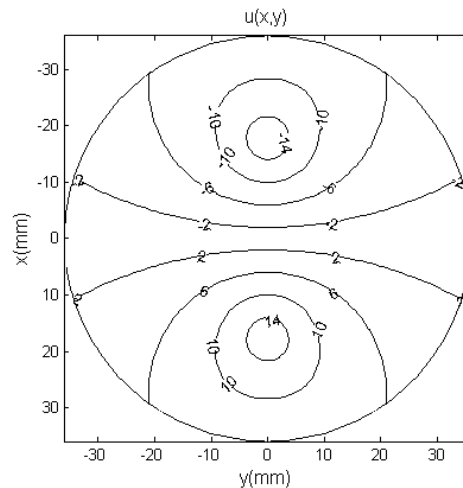


Fig. 2. The level curves obtained by analytically solving the Laplace equation.

The vector height $z(x,y)$ is obtained by a series of equations. Based on the elementary differential geometry, the focal power and astigmatism of the progressive surface are calculated. The contours of them are shown in Fig. 3. The length of progressive corridor is about 16 mm. The width of clear vision area (astigmatism <0.5 diopter) in the distance area at $x = -10$ mm is about 26 mm which is not wide enough for outdoor vision.

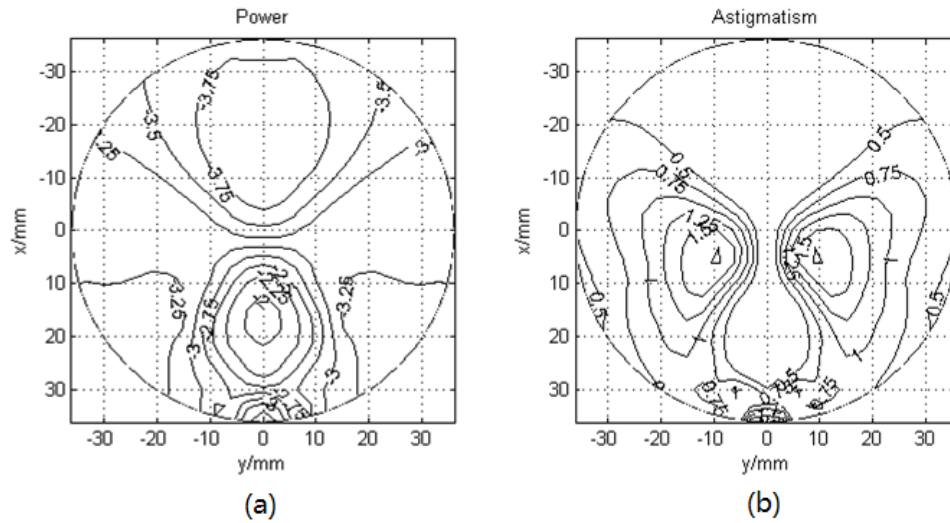


Fig. 3. The focal power (a) and astigmatism (b) of the progressive surface by Winthrop method.

To get a wider distance area, the weighting factor α_i of the objective function to determine the boundary conditions of the Laplace equation is selected as shown in Table 1. The boundary conditions obtained with the genetic algorithm are shown in Fig. 4 and Fig. 5.

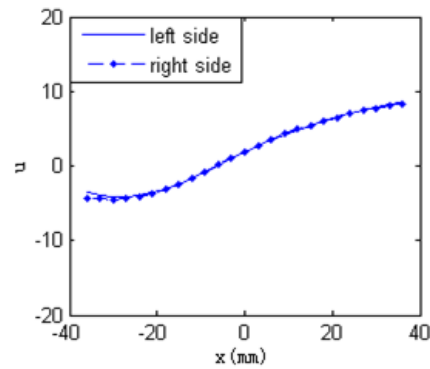


Fig. 4. Boundary conditions of the left and right sides.

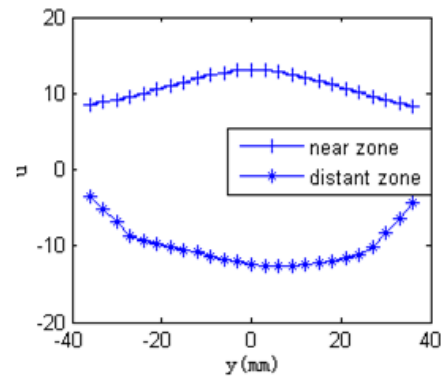


Fig. 5. Boundary conditions of the distance and near zones.

By solving the Laplace equation numerically with the boundary and link conditions, the optimized $u(x, y)$ is obtained. The contours of optimized $u(x, y)$ are shown in Fig. 6. Compare with Fig. 2, the area is wider in which the value of $u(x, y)$ that is smaller than -14 .

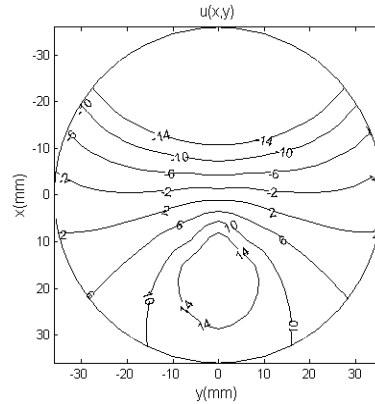
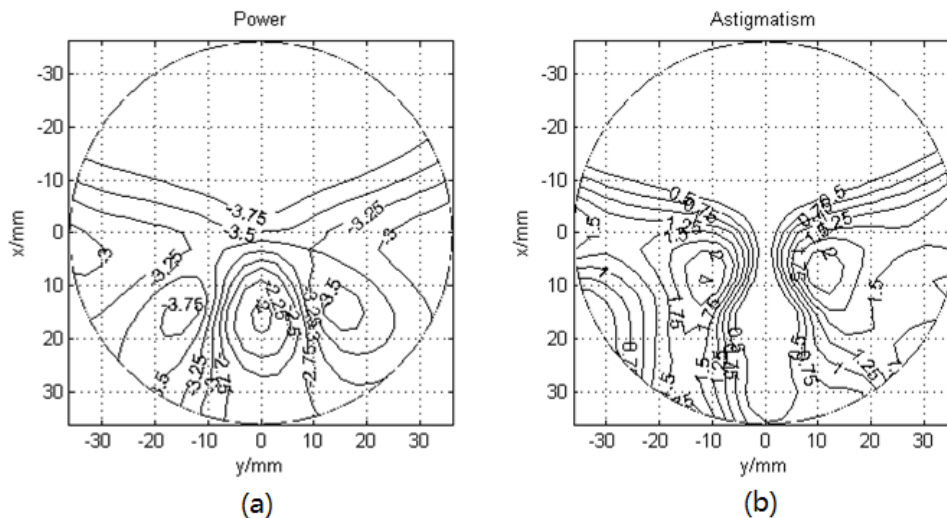


Fig. 6. Contour lines of optimized $u(x, y)$ in the first example.

Once $u(x, y)$ is obtained, $z(x, y)$ can be derived using the above design steps. The contours of the focal power and astigmatism are shown in Fig. 7. The optical performance of the progressive surface is given in Table 3. One can see that the distance area (focal power < -3.75 diopter) in Fig. 7 (a) is greatly improved than that in Fig. 3 (a). The width of the clear vision area (astigmatism < 0.5 diopter) in distance area at $x = -10$ mm is about 46 mm which is more suitable for outdoor vision.



astigmatism is less than 0.02 diopter. Due to the influence of the curvature of the front surface, the width is reduced 12 mm and 2 mm in the distance zone (astigmatism < 0.5 diopter, $x = -10$ mm) and near zone (astigmatism < 0.5 diopter, $x = 18$ mm) of the manufactured PAL than those of the progressive surface.

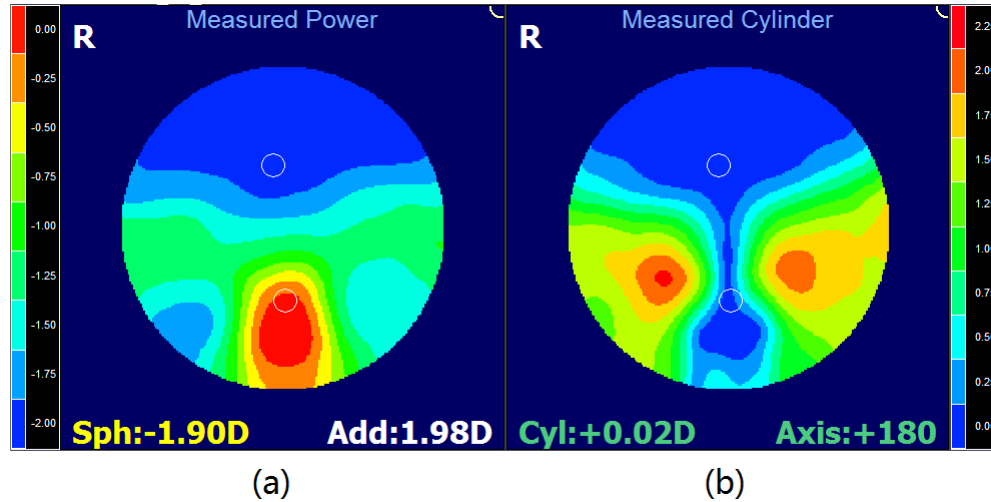


Fig. 8. The focal power (a) and astigmatism (b) of the PAL tested by FFV.

In the second example, the basic parameters are the same as those of the first one. The PAL is used in the office. Therefore, a larger near area and wider corridor are needed. The width d is set to be 9 mm instead of 6 mm as in the first example. The weighting factors based on the need for near vision are shown in Table 2. The boundary conditions obtained with the genetic algorithm are shown in Fig. 9 and Fig. 10. The contours of optimized $u(x, y)$ are shown in Fig. 11.

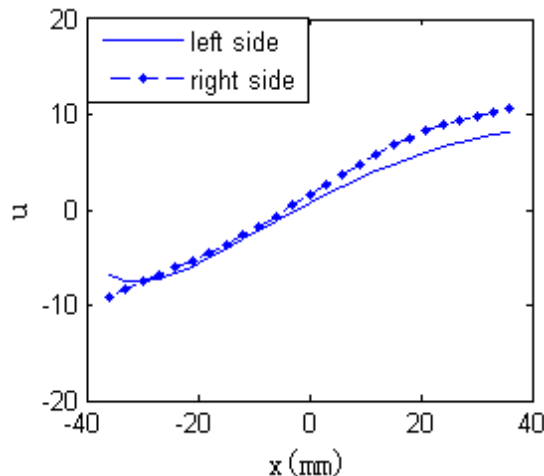


Fig. 9. Boundary conditions of the left and right sides.

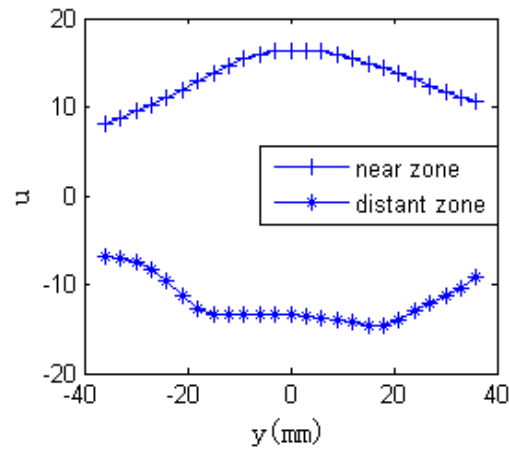


Fig. 10. Boundary conditions of the distance and near zones.

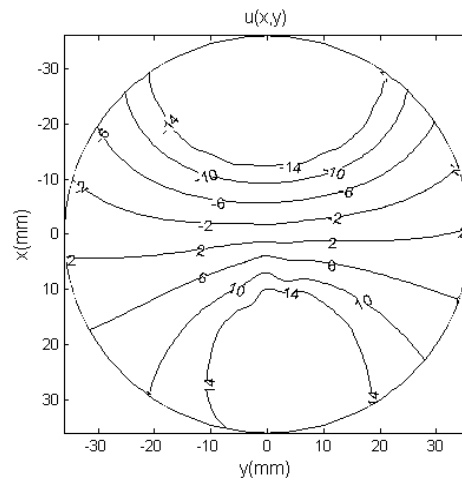
Fig. 11. Contour lines of optimized $u(x, y)$ in the second example.

Figure 12 shows the contours of the focal power and astigmatism of the second example. Table 3 is the optical performance comparison between the first example and the second example. The width of distance area of the first example is 24 mm wider than that of the second example at $x = -10$ mm. The width of near area of the second example is 8 mm wider than that of the first example at $x = 18$ mm. The maximum astigmatism of the second example is smaller than that of the first example, and the width of the corridor is wider.

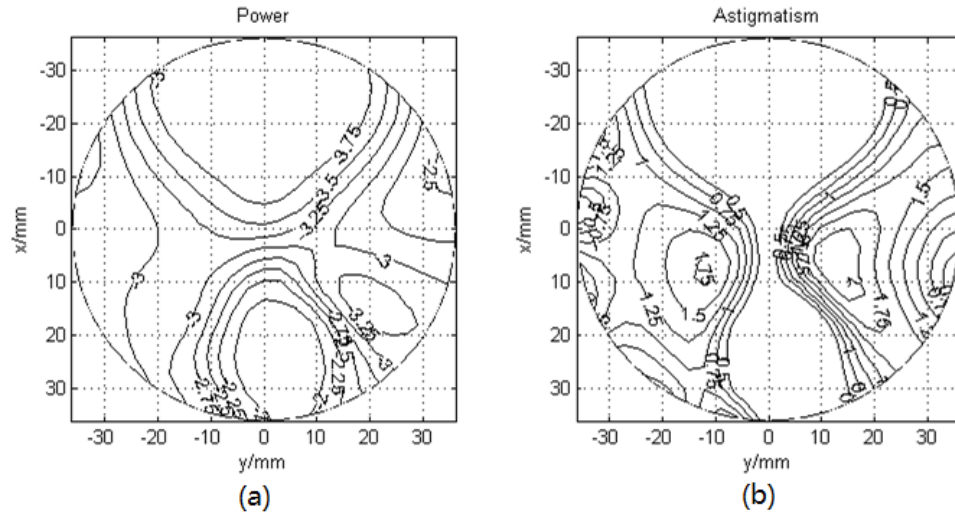


Fig. 12. The focal power (a) and astigmatism (b) of the progressive surface in the second example.

Table 1 and Table 2 are the weighting factors based on the different needs of the wearer. The parameters of ρ and α_i of the objective function are determined by the needs and the preference of the wearer. The astigmatism weighting factor α_2 is selected a larger value for outdoor activities. Larger values of the weighting factors α_3 , α_4 , α_5 and α_6 are selected for office use.

Table 1. The weighting factor of the objective function f in the Eq. (3) for the first example

Parameters	Maximum astigmatism	Distance area	Near area	Progressive corridor	Left astigmatism area	Right astigmatism area
weighting factor of astigmatism($\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6$)	1.2	0.6	0.4	0.8	0.4	0.6
weighting factor of focal power difference($\alpha_7, \alpha_8, \alpha_9$)	-	0.3	0.4	0.2	-	-

Table 2. The weighting factor of the objective function f in the Eq. (3) for the second example

Parameters	Maximum astigmatism	Distance area	Near area	Progressive corridor	Left astigmatism area	Right astigmatism area
weighting factor of astigmatism($\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6$)	1.2	0.1	1.2	0.8	0.8	0.6
weighting factor of focal power difference($\alpha_7, \alpha_8, \alpha_9$)	-	0.1	0.6	0.2	-	-

Table 3. The optical performance in the examples (D is the abbreviation for “diopter”)

	Width of distance area x = −10(astigmatism m<0.5D)/mm	Width of near area x = 18 (astigmatism <0.5D)/mm	Maximum astigmatism /D	Width of corridor x = 6(astigmatism <0.5D)/mm
Analytical method	26	16	1.76	4
The first example	48	10	2.27	3
FFV test in the first example	36	8	2.25	3
The second example	24	18	2.18	4

4. Conclusion

In this study, we have developed a novel design approach that has more control on the auxiliary function and hence meets more individualized vision correction. To accomplish the aim, we solve the Laplace equation numerically. The boundary and link conditions are set to satisfy specific requirements. As a result, a specific need for the dimensions and focal powers of the distance and near regions can be met in the PAL design to a larger extent. The sizes and distributions of the astigmatism areas are also improved with our approach. The examples demonstrate the capability of our approach.

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